QSCDNet: A Hybrid Quantum Spectral Change Detection Network for Hyperspectral Image Change Detection

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***Abstract*—Hyperspectral image change detection (HSI-CD) is an important remote sensing technique for identifying fine-grained land-cover change. Deep learning methods such as convolutional neural networks (CNNs) and transformers have achieved good performance in HSI-CD. However, due to the pseudo-changes caused by imaging conditions, the spectral change characteristics within each pixel often exhibit uncertainties. In this study, differing from the traditional deep learning methods, we aimed to relate the abovementioned spectral change uncertainty to the perspective of the quantum state and built a dual-branch hybrid quantum neural network for HSI-CD (QSCDNet). The quantum branch consists of several 2-D quantum spectral change convolutional blocks (QSCCBs). These blocks provide a wider variety of expression forms of the spectral change, independent of the pseudo-change impact, through a parameterized quantum circuit (PQC), to better extract the change information of the spectral features. The CNN branch is designed based on a channel attention network architecture to provide traditional network change features. The output features of the quantum network branch and the CNN branch are then fused based on a cross-domain change feature fusion module (CDFM). The impact of the number of QSCCBs was also analyzed to verify the effectiveness of increasing the depth of the quantum network structure. The proposed method was tested on three public HSI-CD datasets and compared with the state-of-the-art methods to validate its potential in the field of HSI-CD.**

***Index Terms*—Change detection, hyperspectral image (HIS), quantum neural networks (QNNs), remote sensing.**

I. Introduction

HYPERSPECTRAL image change detection (HSI-CD) is an important technique that utilizes bitemporal hyperspectral images (HSIs) to identify fine-grained land-cover change [1]. HSI-CD can be used in various fields, including natural disaster detection [2], urbanization monitoring [3], and environmental monitoring [4].

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The traditional HSI-CD methods combine change feature extraction with machine learning methods, primarily addressing the challenges of mixed pixels [5], high dimensionality [6], and limited samples [7]. Deep learning methods have attracted much attention in the field of HSI-CD, due to their advantages in high-dimensional feature extraction [8]. For example, Siamese [9], [10], and dual-branch [11] convolutional neural network (CNN) structures have been used in HSI-CD to enhance the change information. Attention mechanisms can also be introduced in HSI-CD networks to focus on the key change features. Band selection combined with a CNN is another way to explore the change information in bitemporal HSIs [12]. Generative adversarial networks (GANs) have also been used in HSI-CD, which employ adversarial training to generate more change samples to consider the class imbalance problem [13]. To better model the spectral contextual information in the HSI-CD task, recurrent neural networks (RNNs) [14] and long short-term memory (LSTM) networks [15] have also been introduced. To further explore the global spectral contextual information, transformer networks based on the multihead self-attention (MHSA) mechanism have been introduced [16], [17]. Researchers have also explored hybrid CNN-transformer architectures to combine both local and global spectral change information, to improve the CD accuracy [18], [19].

However, due to the impact of the imaging conditions, such as the atmosphere, light, angle, and the interference of the spectra from adjacent objects, the spectral features in the original HSI cannot represent the pure reflectivity of the ground target. The traditional deep neural networks have some limitations in explicitly modeling the abovementioned spectral interference factors, and the CD accuracy can be affected.

Inspired by quantum computing [20], [21], [22], which has advantages in parallel computing [23] and feature expression [24], we consider the abovementioned problem from the perspective of quantum computing. The HSI information is encoded into a quantum state to extract the change features through a parameterized quantum circuit (PQC) [25]. Finally, we propose a deep quantum neural network (QNN) architecture for the HSI-CD task.

QNNs have gradually received attention in the image processing field. Scholars have utilized PQCs to design quantum convolution modules and construct the network architecture. This approach has demonstrated the feature extraction ability

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of quantum convolution in some image classification tasks [24], [26]. Meanwhile, the QNN architecture has also begun to receive attention in the field of remote sensing, in applications such as scene classification [27], [28], HSI restoration [29], and HSI-CD [30].

In this study, differing from the abovementioned works, we further studied the use of the QNN architecture for HSI processing and proposed to model the spectral change information in hyperspectral imagery by the quantum state. We then constructed a quantum network for the HSI-CD task, which has the potential to reduce the impact of the potential interference factors by optimizing the angle of the quantum gates in the quantum circuit during the network training process. The major contributions of this article are as follows.

1) A dual-branch hybrid QNN (QSCDNet) for HSI-CD is proposed, where the quantum network branch is combined with a traditional CNN branch to enhance the presentation of HSI change features and suppress the false changes.
2) The spectral value in the HSI is assumed to be a mixture of the pure reflectivity and the interference factors, and is modeled as the quantum superposition state to construct a quantum network branch for HSI-CD. This reduces the spectral uncertainties by optimizing the angle information of the quantum gates in the quantum circuit during the network training process.
3) The proposed quantum network branch is designed based on 2-D quantum spectral change convolutional blocks (QSCCBs), which apply the PQC on the cross-temporal 2-D spectral matrix to acquire the hyperspectral change features. The advantage of the quantum network branch lies in the use of quantum entanglement to express the spectral changes between different phases.
4) The impact of the depth of the QNN was studied, and a quantum residual network structure was designed to address the "quantum barren plateau" problem in change detection.

The rest of this article is organized as follows. In Section II, the related CNN and transformer methods in HSI-CD are introduced, and we survey the progress of QNNs in remote sensing image processing. In Section III, we introduce the proposed method in detail. Section IV describes the experimental settings, Section V gives the experimental results and analysis, and Section VI provides our conclusion.

II. Releted Work

A. CNN-Based HSI-CD Methods

Wang et al. [31] proposed an end-to-end 2-D CNN method that leverages a hybrid affinity matrix to provide rich cross-channel gradient information. Mou et al. [32] extracted bitemporal hyperspectral change features using a 2-D CNN and subsequently employed an RNN to establish the temporal correlation. Lin et al. [33] employed a Siamese CNN to enhance the bitemporal features and used a linear outer product to enhance the change information. Zhan et al. [34] focused on both the spectral and spatial bitemporal change features by using 1-D and 2-D convolution operations. Qu et al. [35] combined a CNN and LSTM network to capture spatial–spectral–temporal change information. Shi et al. [15] constructed a Siamese CNN-LSTM network to learn the multiscale temporal characteristics of bitemporal HSIs. Wang et al. [9] introduced a spatial–spectral attention mechanism into the Siamese CNN encoder to highlight the spectral bands and spatial locations which contain critical change information. Song et al. [36] employed a spatial–spectral attention mechanism to better discriminate the key change features. Dong et al. [37] proposed a cross-attention mechanism to enhance the change information. Wang et al. [38] proposed a spatial–spectral attention based feature fusion mechanism to combine features from different layers, strengthening the effective semantic change information. Zhang et al. [39] introduced a multiscale cross-attention mechanism to extract more discriminative change information from multilevel deep features. Li et al. [40] proposed a 2-D spatial–spectral self-attention module to enhance the spectral differences. Ou et al. [41] cropped multiscale patches from bitemporal HSIs to further improve the spatial change information. Wang et al. [42] employed gated recurrent units to extract multilevel change information, to improve the accuracy of the HSI-CD network.

B. Transformer-Based HSI-CD Methods

Transformer models [43], which are based on the MHSA mechanism, have also been studied in HSI-CD [44], due to their ability to model long-range dependencies. Wu et al. [45] performed feature fusion using the MHSA mechanism to highlight the change features. Wang et al. [46] employed a transformer model to separately extract spectral and spatial features from bitemporal HSIs. Zhang et al. [47] used a transformer model as the backbone and applied both differential and dot-product operations on the deep features to extract more discriminative change information. Zhang et al. [48] used a cascaded transformer structure to extract spatial and spectral change information. Ding et al. [49] proposed an HSI-CD network based on a Siamese structure, where multiscale features extracted by the Siamese CNN are input into the transformer to aggregate fine-grained details and high-level semantics. Dong et al. [8] introduced a hybrid CNN-transformer structure to jointly extract local and global features and enhance the generalization ability in complex scenarios. Chen et al. [50] modified the MHSA mechanism by incorporating cross-attention in some of the attention heads.

C. QNN in Remote Sensing Image Processing

Recently, quantum computing has begun to receive attention in machine learning due to its high parallelism characteristic, to accelerate the efficiency of the algorithms. The superposition state and quantum entanglement have also been proven to be effective in improving the accuracy in certain machine learning applications [21]. The QNN developed by Gupta and Zia [22] was the first quantum-based neural network, which utilizes qubits and quantum gates to construct the network architecture, and its effectiveness was demonstrated

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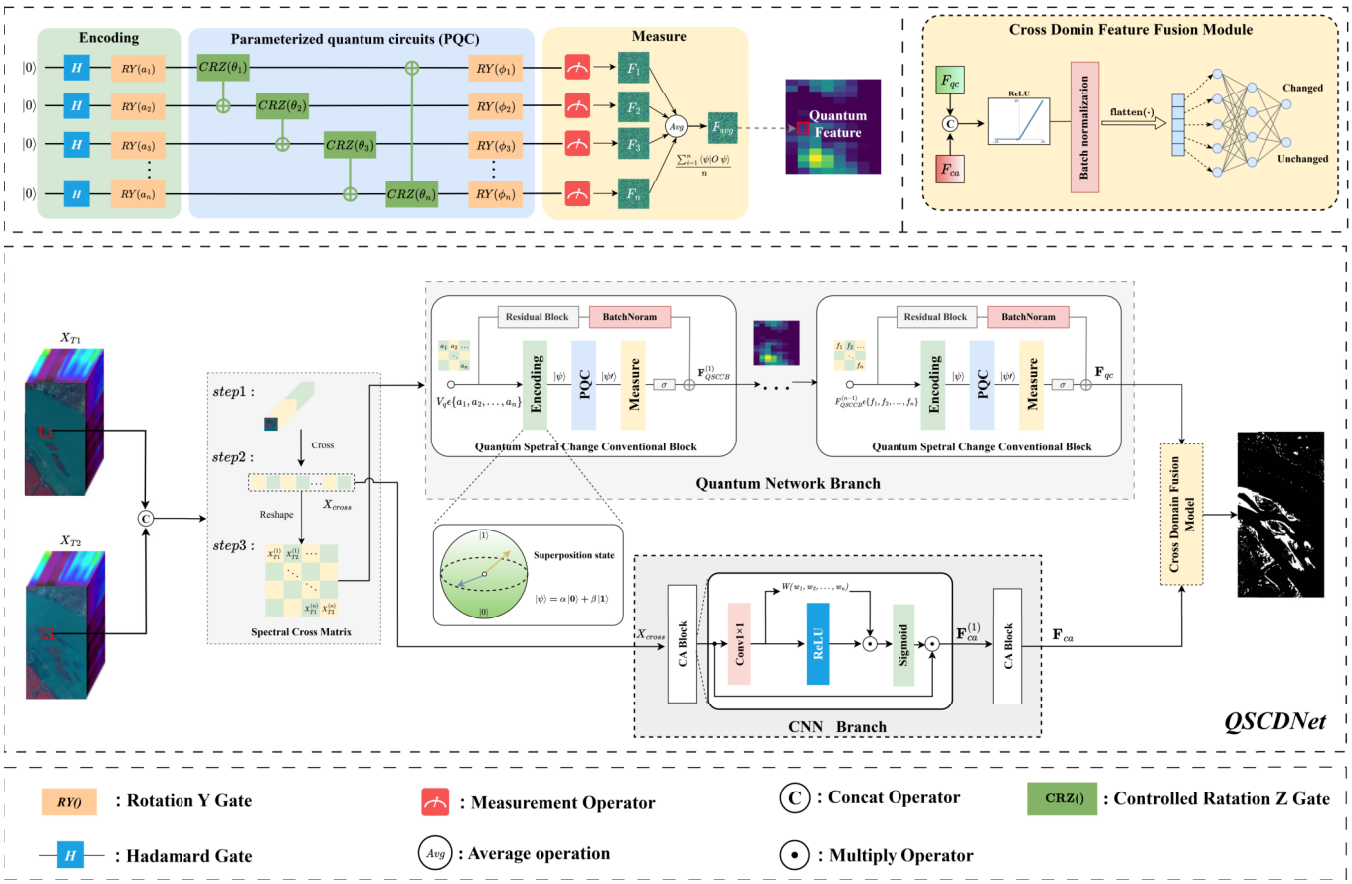


Fig. 1. Network structure of QSCDNet consists of two branches and a CDFM: the upper branch is the quantum network branch, which extracts quantum features of the imagery by inputting traditional features into quantum circuits. The branch below is the CNN branch, which is mainly provided with classical features by a weighted channel attention network. On the far right is the CDFM, which is used to combine the quantum features with traditional features.

through experiments. In the field of image processing, the quantum convolutional neural network (QCNN) is a representative image classification method [51]. By introducing quantum superposition states and entanglement characteristics into a traditional CNN, the model can simultaneously consider multiple possibilities of the objective, thereby improving its expressive and generalization capabilities [23]. Li et al. [52] achieved a deeper QCNN by using several consecutive quantum convolutional layers and one quantum classifier layer. Bokhan et al. [53] proposed a quantum method based on the QCNN architecture to solve multiclass classification problems. Meanwhile, QNNs have also been studied in the field of remote sensing image processing. Sebastianelli et al. [26] proposed a remote sensing image classification model that incorporates quantum layers into a classical neural network. Lin and Chen [29] proposed a hyperspectral quantum deep network that encodes the hyperspectral information using a small number of qubits and learns the mapping from measurement statistics to target imagery. Fan et al. [28] proposed two hybrid quantum-classical deep learning frameworks for multispectral remote sensing image classification. Both models extract features from multispectral images using quantum computing and perform the final classification using classical computing. Lin et al. [30] combined a quantum network with a graph neural network (GNN) and designed a quantum classifier for HSI-CD.

III. Proposed Method

The structure of the proposed QSCDNet is displayed in Fig. 1. The network follows a dual-branch structure, including a deep quantum network branch and a CNN branch. The deep quantum network branch implements the aforementioned quantum circuit based on a PQC [25], which consists of several 2-D QSCCBs. These blocks map the bitemporal spectral data of each pixel into a cross-temporal 2-D spectral matrix, and the variational layer in quantum convolution can more effectively extract the change information from this matrix. The quantum state helps to consider the spectral uncertainty caused by the interference factors, and the use of the quantum gates can increase the sensitivity of the network to subtle spectral differences. In order to enhance the diversity of the change features while taking into account the classical spectral change information, a classical CNN branch based on a channel attention network architecture, inspired by SENet [54], is also introduced. A cross-domain change feature fusion module (CDFM) is then introduced to further extract the key change information from both the quantum network branch and the CNN branch. Finally, the fused change features are connected with the classifier layer to acquire the binary change map. In addition, in the PQC, a controlled RZ gate (*CRZ* gate) with angle parameters is employed to establish the entanglement correlations between adjacent qubits.

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A. Quantum Network Branch and CNN Branch

The input bitemporal HSI images are represented as $X_{T1}, X_{T2} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the height, width, and number of spectral bands. For each pixel, the band values at different time phases are alternately arranged to acquire a cross-temporal feature vector X_{cross}, as shown in (1). Subsequently, X_{cross} is reshaped into a cross-temporal 2-D spectral matrix A_{spectral} as the input of the deep quantum network branch. The alternating arrangement strategy for the spectra at different time phases helps to enhance the change information by considering the long-range spectral dependencies for each pixel

$$X_{\text{cross}} = \left\{X_{T1}^{(1)}, X_{T2}^{(1)}, \ldots, X_{T1}^{(C)}, X_{T2}^{(C)}\right\} \tag{1}$$

$$A_{\text{spectral}} = \text{Reshape}\left(X_{\text{cross}}\right) \tag{2}$$

where Reshape$(\cdot)$ represents reshape operation to reshape the spectral sequence into a 2-D spectral matrix.

As shown in Fig. 1, the quantum network branch is composed of several QSCCBs. Each QSCCB is designed as a residual quantum convolutional structure, according to the following equation:

$$\mathbf{F}_{\text{QSCCB}}^{(n)} = \sigma\left(\left(f_{Q\text{Conv}}^{K\times K}\left(F_{\text{QSCCB}}^{(n-1)}\right)\right) + \text{BN}\left(f_{\text{Conv}}^{3\times 3}\left(F_{\text{QSCCB}}^{(n-1)}\right)\right)\right) \tag{3}$$

where $\mathbf{F}_{\text{QSCCB}}^{(n-1)}$ expresses the output feature of the $(n\text{-}1)$th block, which is used as the input of the nth block. Note that A_{spectral} is the input of the first block. The residual structure is achieved by combining the output features of the quantum convolution [23] with the sum of the features extracted by a traditional 3×3 convolution operation. K represents the size of the quantum convolution kernel, BN represents the batch normalization layer, and σ represents the rectified linear unit (ReLU) activation function.

First, the input feature $\mathbf{F}_{\text{QSCCB}}^{(n-1)}$ is flattened into a feature vector $V_q \in \{a_1, a_2, a_3, \ldots, a_i\}$. Each element a_i in V_q is encoded into the Hilbert space and acquires the quantum superposition state $|\psi_i\rangle$ based on the following equation:

$$|\psi_i\rangle = H \cdot RY\left(a_i\right)|0\rangle \tag{4}$$

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \tag{5}$$

$$RY\left(\theta\right) = \begin{pmatrix}\cos\frac{\theta}{2} & -\sin\frac{\theta}{2}\\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{pmatrix} \tag{6}$$

where H is the Hadamard gate [55], which encodes the ground state $|0\rangle$ for each quantum bit into the superposition state. $RY(\cdot)$ is the rotating-Y gate [55], where a_i plays the role of the rotation parameter θ of the state vector of the qubit around the y-axis on a Bloch sphere [56].

Next, quantum convolution $f_{Q\text{Conv}}^{K\times K}$ is performed on the superposition state $|\psi_i\rangle$ by the PQC, which consists of an RY gate and a CRZ gate [55]. The RY gate is a single-bit quantum gate used to extract the features of the qubits themselves. The CRZ gate is a dual-bit quantum gate used to establish the interaction between two qubits. Due to the temporal cross-arrangement of spectral features A_{spectral}, the CRZ gate can be used to extract the change information. For a K^2-bit quantum system, the entangled quantum state $|\psi'\rangle$ can be obtained by the following equation:

$$\left|\psi'\right\rangle = \bigotimes_{i=1}^{K^2} RY\left(\theta_i\right) \cdot \prod_{j=1}^{K^2-1} CRZ_{j,j+1}\left(\phi_j\right)|\psi\rangle \tag{7}$$

$$CRZ\left(\theta\right) = \begin{pmatrix}I & 0\\ 0 & e^{i\theta}\end{pmatrix} \tag{8}$$

where θ and ϕ are the parameters of the quantum gates, $\otimes$ represents the tensor product, and $\prod$ represents the continuous production of the quantum gates. The introduction of the superposition state assumes the impact of the potential interference factors to be the rotation parameters of the RY gate, which helps to reduce the spectral uncertainty during the back-propagation process. The length of the data input into the quantum circuit is K^2.

For a certain quantum state $|\psi'_i\rangle$ in the system, it can be obtained by taking partial traces of the density matrix

$$\left|\psi'_i\right\rangle = \mathbf{Tr}_{\neq i}\left(\rho\right) \tag{9}$$

$$\rho = \left|\psi'\right\rangle\left\langle\psi'\right| \tag{10}$$

where $|\psi'_i\rangle$ is a single quantum state in the system, and $\mathbf{Tr}(\cdot)$ represents the trace operation. $\boldsymbol{\rho}$ is the density matrix obtained by the outer product of the state vector.

In order to combine the quantum features with a traditional neural network, the quantum state should be converted into a certain classical value. The measured features $\mathbf{F}_{qc}$ from the quantum circuit can be acquired as follows:

$$\mathbf{F}_{qc} = \frac{\sum_{i=1}^{n} Re\left(\left\langle\psi'_i\right|\hat{O}_i\left|\psi_i{}'\right\rangle\right)}{n} \tag{11}$$

where $\hat{O}_i$ stands for the measurement-related operator, and $\langle\psi_i|$ denotes the conjugate transpose of the quantum state. $Re(\cdot)$ indicates the calculation expectation.

An extra classical CNN branch based on a channel attention network architecture is introduced. As an auxiliary branch of the quantum network branch, it provides classical feature information to further improve the CD accuracy. By introducing a channel attention network, the channel features can be adaptively enhanced to obtain the key channel features $\mathbf{F}_{ca}$

$$\mathbf{F}_{ca}^{(1)} = \mathbf{X}_{\text{cross}} \cdot \left(\sigma\left(W_1 \cdot \text{Conv1D}\left(X_{\text{cross}}\right)\right)\right) \tag{12}$$

$$\mathbf{F}_{ca} = \mathbf{F}_{ca}^{(1)} \cdot \left(\sigma\left(W_2 \cdot \text{Conv1D}\left(\mathbf{F}_{ca}^{(1)}\right)\right)\right) \tag{13}$$

where W_1 and W_2 are trainable weights, σ is the ReLU activation function, and Conv1D$(\cdot)$ represents 1-D convolution. $\mathbf{F}_{ca}^{(1)}, \mathbf{F}_{ca}$ represent the first and second channel attention features, respectively, where $\mathbf{X}_{\text{cross}}$ is the input to the first module, and $\mathbf{F}_{ca}^{(1)}$ is the input to the second module.

B. Cross-Domain Change Feature Fusion Module

In order to better fuse the features of the dual branches, the CDFM is introduced to aggregate the dual-branch features

$$\mathbf{F}_{\text{fuse}} = \text{BN}\left(\sigma\left(\text{Concat}\left[\mathbf{F}_{ca}, \mathbf{F}_{qc}\right]\right)\right) \tag{14}$$

where $\mathbf{F}_{ca}$ represents the output feature from the classical CNN branch, and $\mathbf{F}_{qc}$ represents the quantum spectral feature extracted by the quantum network branch. Concat$(\cdot)$ represents

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the concatenation operation. The fused feature $\mathbf{F}_{\text{fuse}}$ is obtained by the activation function and batch normalization.

At the end of the network, three fully connected layers followed by a softmax layer are applied to convert the features into probability rules to distinguish the changed and unchanged types

$$\text{softmax}(x) = \frac{e^{x_i}}{\sum_i e^{x_i}} \tag{15}$$

where i represents the number of classes, which is 2 in this article.

In the proposed approach, the PQC is combined with the classical deep neural network structure, and optimization of the quantum gate parameters is achieved by backpropagation. This approach enables the training of a PQC on a classical computer while keeping the advantage of quantum computing.

IV. Experimental Settings

A. Descriptions of the Datasets

Three public HSI-CD datasets—River [31], Hermiston, and Farmland [57]—were used in the experiments. All the datasets were obtained by the Hyperion sensor onboard the Earth Observing-1 (EO-1) satellite [40]. The Hyperion sensor features 242 bands covering the spectral range from 400 to 2500 nm, with a spatial resolution of 10 m and a spectral resolution of 30 nm [58]. Fig. 2 presents false-color composite images (using bands 39, 19, and 9 as the RGB bands [48]) of the three datasets.

1) River Dataset: The River dataset comprises bitemporal images captured in Jiangsu province, China, on May 3, 2013, and December 31, 2013. The dataset has a spatial size of 463 × 241 pixels and includes 198 spectral bands. The major changed areas are caused by the reduction of the river channel. The numbers of changed and unchanged pixels are 9 698 and 1 01 885, respectively.

2) Hermiston Dataset: The Hermiston dataset was collected in Hermiston, Umatilla County, USA, with the bitemporal images captured on May 1, 2004, and May 8, 2007. The dataset has a spatial size of 390 × 200 pixels and 242 spectral bands. The major changed areas are caused by changes in the water content of the farmland. The numbers of changed and unchanged pixels are 19 972 and 58 028, respectively.

3) Farmland Dataset: The Farmland dataset consists of bitemporal images from a wetland agricultural region in Yancheng, Jiangsu province, China, taken on May 3, 2006, and April 23, 2007. This dataset has a spatial size of 450 × 140 pixels and 155 spectral bands. The numbers of changed and unchanged pixels are 18 277 and 44 723, respectively.

B. Implementation Detials

1) Training Settings: The experiments presented in this article were conducted on a workstation equipped with a NVIDIA GeForce 3090 GPU and implemented in a Python environment. The quantum platform used in the experiments was TensorCircuit. The same parameters were applied across all three datasets. The batch size was set to 512, the training epochs were set to 180, and the initial learning rate was set to 0.001. The Adam optimizer was used, and a cross-entropy loss function was adopted. The number of qubits was selected as nine, according to the ablation study.

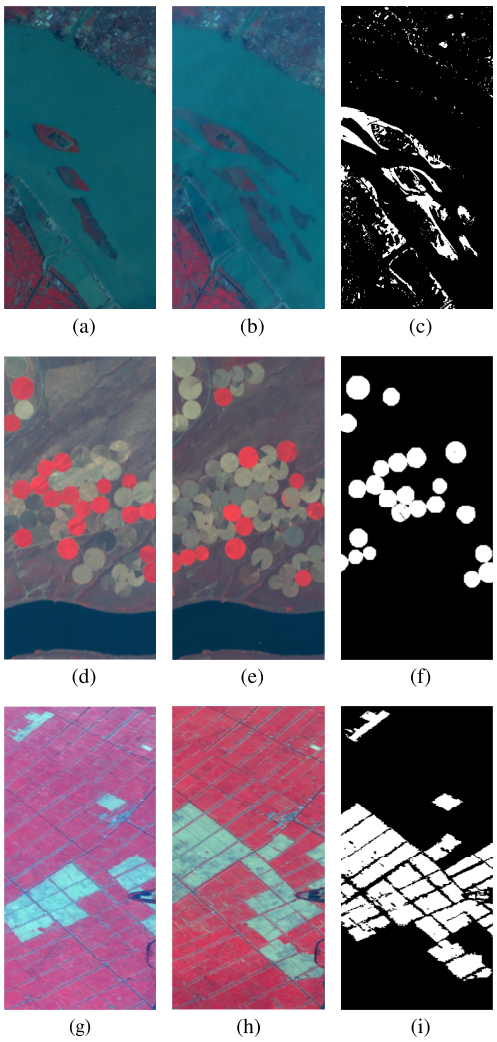


Fig. 2. False-color images of the three datasets. (a) River dataset on May 3, 2013. (b) River dataset on December 31, 2013. (c) Ground-truth labels for the River dataset. (d) Hermiston dataset on May 1, 2004. (e) Hermiston dataset on May 8, 2007. (f) Ground-truth labels for the Hermiston dataset. (g) Farmland dataset on May 3, 2006. (h) Farmland dataset on April 23, 2007. (i) Ground-truth labels for the Farmland dataset.

C. Evaluation Metrics

The overall accuracy (OA), kappa coefficient (kappa), precision (P), recall (R), and $F1$-score ($F1$) of the changed class are used as the indicators to evaluate the effectiveness of QSCDNet.

The OA measures the proportion of both the correctly classified changed and unchanged pixels, and is calculated as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{16}$$

where TP, TN, FP, and FN are the number of true positives, true negatives, false positives, and false negatives, respectively.

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TABLE I
COMPARATIVE EXPERIMENTAL RESULTS FOR DIFFERENT NUMBERS OF QSCCBS WITH 1% TRAINING SAMPLES ON THE RIVER DATASET AND FARMLAND DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

Dataset	Method	Blocks	OA (%)	Kappa (%)	F1 (%)	P (%)	R (%)
River	QSCDNet	1	96.34	74.94	76.90	**85.50**	69.89
		2	96.35	77.96	79.97	76.12	**84.12**
		4	**96.79**	**79.94**	**81.70**	81.12	82.32
Farmland	QSCDNet	1	97.46	93.74	95.51	97.80	93.33
		2	97.77	94.57	96.14	96.85	95.44
		4	**98.15**	**95.47**	**96.77**	**97.83**	**95.73**

Kappa is calculated as follows:

$$\text{Kappa} = \frac{\text{OA} - P_e}{1 - P_e} \tag{17}$$

$$P_e = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{FN} + \text{TN})(\text{FP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{18}$$

The $F1$-score is the harmonic mean of the precision and recall, offering a more balanced evaluation of the prediction accuracy

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{19}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{21}$$

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Parameter Analysis and Ablation Study

1) Parameter Analysis: The effect of the number of QSCCBs on the accuracy for a single quantum network branch was analyzed on the River and Farmland datasets. The results are listed in Table I.

For the River dataset, the OA, kappa, and $F1$-score values improve with the increase of the number of QSCCBs. When only one QSCCB is used, the OA, kappa, and $F1$-score values are 96.34%, 74.94%, and 76.90%, respectively. When using two blocks, the OA, kappa, and $F1$-score values are increased by 0.01%, 3.02%, and 3.07%, respectively, compared with the result of using only one block. The highest accuracy is achieved when four blocks are used, reaching 96.79%, 79.94%, and 81.70% in OA, kappa, and $F1$-score, respectively. For the precision, one QSCCB obtains the highest accuracy, reaching 85.50%. The accuracy for two blocks and four blocks is decreased to 84.12% and 82.32%, respectively. The best recall is reached when the number of blocks is two. However, the highest value of $F1$-score indicates that four blocks show the best performance in keeping the balance between false alarms and missed detections.

For the Farmland dataset, the OA, kappa, $F1$-score, and recall values increase with the increase in the number of blocks. Four blocks result in the highest values, of 98.15%, 95.47%, 96.77%, and 97.83% in OA, kappa, $F1$-score, and recall, respectively. For the precision, the accuracy for two blocks is decreased when compared with that for one block, but four blocks still result in the highest accuracy, at 95.73%.

2) Ablation Study of the Proposed Quantum Network Branch: The effectiveness of the quantum network branch in QSCDNet was analyzed on the three datasets. The results are presented in Table II.

For the River dataset, the first row shows the result of the single-branch network based on the traditional channel attention network architecture, the second row shows the result of using only the quantum network branch, and the third row shows the result of the complete QSCDNet. For the traditional CNN branch, the OA, kappa, $F1$-score, precision, and recall values are 94.63%, 60.86%, 63.66%, 80.28%, and 53.92%, respectively. The quantum network branch results in values of 96.40%, 76.79%, 78.70%, 80.91%, and 76.63%, respectively, on the five corresponding indicators, representing improvements of 1.77%, 15.93%, 15.04%, 0.63%, and 22.71%, respectively. The best performance is acquired when both branches are used, resulting in values of 96.79%, 79.94%, 81.70%, 81.12%, and 82.32%, respectively, for the five indicators.

For the Hermiston dataset, using only the quantum network branch yields OA, kappa, $F1$-score, and precision values of 98.60%, 94.08%, 94.82%, and 94.78%, respectively, which are 0.34%, 1.87%, 1.50%, and 3.42% higher than the results of the traditional CNN. The recall is 94.67%, which is 0.46% lower than that of the traditional CNN. The complete QSCDNet obtains the highest values of the five indicators, at 98.64%, 95.05%, 96.48%, 96.57%, and 96.41%, respectively.

For the Farmland dataset, the dual-branch network acquires the highest OA, kappa, $F1$-score, and precision values, of 97.73%, 94.49%, 96.09%, and 95.81%, respectively. For the recall, it is 0.33% lower than that of the single-branch QCNN.

It is worth noting that, for all three datasets, the addition of the quantum network branch results in a significant improvement in $F1$-score, compared with the traditional CNN branch, with improvements of 3.0%, 1.66%, and 0.36%, respectively. This indicates that incorporating quantum features can better balance the accuracy of the changed and unchanged categories.

3) Ablation Study for the Number of Qubits: The performance with four and nine qubits for each QSCCB block was analyzed on the River, Hermiston, and Farmland datasets. The results are presented in Table III.

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TABLE II
QUANTITATIVE RESULTS OF THE ABLATION STUDY RELATED TO THE PROPOSED TECHNIQUES ON THE RIVER, FARMLAND, AND HERMISTON DATASETS

Dataset	Structure	Method	OA (%)	Kappa (%)	F1 (%)	P (%)	R (%)
River	Single Branch	CNN	94.63	60.86	63.66	80.28	53.92
	Single Branch	QCNN	96.40	76.79	78.70	80.91	76.63
	Dual Branch	QSCDNet	**96.79**	**79.94**	**81.70**	**81.12**	**82.32**
Hermiston	Single Branch	CNN	98.26	92.21	93.32	91.36	95.13
	Single Branch	QCNN	98.60	94.08	94.82	94.78	94.67
	Dual Branch	QSCDNet	**98.64**	**95.05**	**96.48**	**96.57**	**96.41**
Farmland	Single Branch	CNN	97.42	93.73	94.93	95.00	94.21
	Single Branch	QCNN	97.50	93.96	95.73	94.86	**96.61**
	Dual Branch	QSCDNet	**97.73**	**94.49**	**96.09**	**95.81**	96.38

TABLE III
QUANTITATIVE RESULTS OF THE ABLATION STUDY FOR NUMBER OF QUBITS IN THE PROPOSED METHOD ON THE RIVER, FARMLAND, AND HERMISTON DATASETS

Dataset	Num_Qubits	OA (%)	Kappa (%)	F1 (%)	P (%)	R (%)
River	4	96.78	79.30	**81.06**	**82.96**	79.24
	9	**96.79**	**79.94**	80.06	81.12	**79.33**
Hermiston	4	**98.77**	94.49	95.19	95.67	94.71
	9	98.64	**95.05**	**96.48**	**96.57**	**96.41**
Farmland	4	97.59	94.47	96.03	95.20	**96.87**
	9	**97.73**	**94.49**	**96.09**	**95.81**	96.38

For the River dataset, when using four qubits, the OA, kappa, $F1$-score, precision, and recall values are 96.78%, 79.30%, 81.06%, 82.96%, and 79.24%, respectively. Compared with the case with four qubits, when using nine qubits, the $F1$-score and precision are decreased by 1.0% and 1.74%, respectively, while the OA, kappa, and recall are increased by 0.01%, 0.64%, and 0.09%, respectively. The decrease in $F1$-score for nine qubits suggests that the use of four qubits in the QSCCB block is more suitable for the River dataset.

For the Hermiston dataset, considering four qubits, the OA, kappa, $F1$-score, precision, and recall are 98.77%, 94.49%, 95.19%, 95.67%, and 94.71%, respectively. The results under nine qubits give 0.56%, 1.29%, 0.90%, and 1.70% increases for the kappa, $F1$-score, precision, and recall, respectively, and a decrease of 0.13% for the OA, compared with the condition under four qubits. These results suggest that the QSCCB block with nine qubits is more suitable for this dataset.

For the Farmland dataset, when using four qubits, the OA, kappa, $F1$-score, precision, and recall are 97.59%, 94.47%, 96.03%, 95.20%, and 96.87%, respectively. Using nine qubits gives a better performance for the OA, kappa, $F1$-score, and precision, with values of 97.73%, 94.49%, 96.09%, and 95.81%, respectively. The recall under nine qubits is 96.38%, which is lower than the condition of four qubits. These results indicate that the QSCCB block with nine qubits is more suitable for the Farmland dataset.

B. Comparison Results and Analysis

To validate the effectiveness of the proposed QSCDNet, several related HSI-CD methods were compared, i.e., the CNN-based methods of convolutional LSTM (ConvLSTM) [15] and the channel spatial attention network (CSANet) [36], the transformer-based methods of SSTFormer [46], the multilevel encoder–decoder attention network (ML-EDAN) [35], the differential information-enhanced feature exchange network (DIEFEN) [45], and the multilevel features fused and change information-enhanced neural network (MFCEN) [49], in addition to the quantum algorithm based method of the quantum information-empowered GNN (QUEEN-G) [30]. We randomly selected 1% of the samples of each dataset as the training samples.

1) Results and Analysis of the River Dataset: Table IV lists the accuracies of the various methods on the River dataset.

For the OA, the accuracies of the CNN-based methods of CSANet and ConvLSTM are 95.80% and 93.36%, respectively. For the transformer-based methods, MFCEN acquires the best OA, of 96.59%. Compared with CSANet, MFCEN shows an improvement in OA of 0.79%. The proposed QSCDNet method obtains the highest OA, of 96.79%, which is 0.20% higher than that of the second-place MFCEN.

For the kappa and $F1$-score, the proposed method also achieves the best results, reaching 79.94% and 80.06% respectively, which are 2.06% and 0.33% higher than the results of the second-best MFCEN. The highest $F1$-score suggests that the use of quantum features can alleviate the imbalance problem between changed and unchanged samples on this dataset. For the precision, there is a significant difference between the different methods. ConvLSTM and QUEEN-G achieve precision values of 62.27% and 60.80%, respectively, and ML-EDAN and DIEFEN achieve precision values of 75.10%

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TABLE IV
COMPARATIVE EXPERIMENTAL RESULTS OF THE DIFFERENT HSI-CD METHODS WITH 1% TRAINING SAMPLES ON THE RIVER DATASET, WITH THE HIGHEST SCORE HIGHLIGHTED IN BOLD RED, THE SECOND SCORE HIGHLIGHTED IN BOLD GREEN, AND THE THIRD SCORE HIGHLIGHTED IN BOLD BLUE

Method	OA (%)	Kappa (%)	F1 (%)	P (%)	R (%)
ConvLSTM	93.36	64.04	67.56	62.27	**77.21**
CSANet	95.80	70.96	73.21	**82.18**	66.06
SSTFormer	95.56	67.19	69.49	82.10	58.75
ML-EDAN	95.12	67.52	70.17	75.10	66.19
MFCEN	**96.59**	**77.88**	**79.73**	**82.92**	76.90
QUEEN-G	94.3	70.6	73.6	60.8	**94.1**
DIEFEN	**96.06**	**74.27**	**76.41**	79.67	73.49
QSCDNet (proposed)	**96.79**	**79.94**	**80.06**	**81.12**	**79.33**

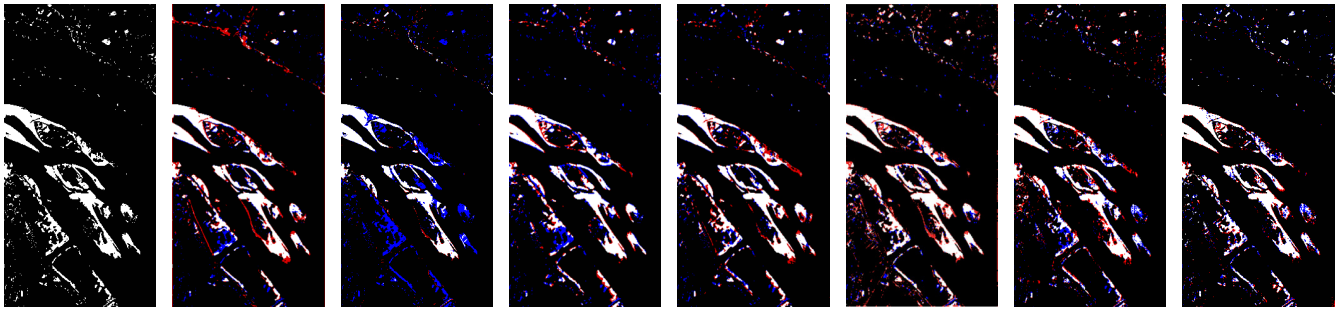


Fig. 3. HSI-CD prediction map of all the methods on the River dataset. (Left to right) For each row, are the ground truth, the ConvLSTM prediction results, the SSTFormer prediction results, the CSANet prediction results, the MFCEN prediction results, the QUEEN-G prediction results, the DIEFEN prediction results, and the QSCDNet prediction results. The false negative and false positive pixels are denoted in red and blue colors, respectively.

and 79.67%, respectively. CSANet, SSTFormer, MFCEN, and QSCDNet all achieve precision values of over 80%, with the highest being MFCEN and the second-highest being CSANet. For the recall, there are also significant differences between the different methods. CSANet, SSTFormer, and ML-EDAN achieve recall values of 66.06%, 58.75%, and 66.19%, respectively. ConvLSTM, MFCEN, QUEEN-G, DIEFEN, and QSCDNet all obtain recall values of over 70%, at 77.21%, 76.90%, 94.10%, 73.49%, and 79.33%, respectively. Although QUEEN-G obtains the highest recall, of 94.10%, the precision is only 60.80%, indicating that this method is not very balanced on this dataset. It is worth noting that the proposed method maintains a good balance on this set of data, with the precision and recall around 80%, and the $F1$-score is the only one that exceeds 80%.

The visual results for the River dataset are shown in Fig. 3. It can be concluded that CSANet, SSTFormer, and ML-EDAN have a significant number of missed detections. ConvLSTM and QUEEN-G obtain a high number of false alarms. The visual results of MFCEN, DIEFEN, and QSCDNet are relatively balanced. The proposed QSCDNet achieves the best visual performance. Compared with the prediction maps of the other HSI-CD methods, the false alarms and false positives of the proposed QSCDNet are significantly lower, indicating that both the missed detection rate and false detection rate of QSCDNet are low, demonstrating the generalization ability of the proposed model. Especially in the sparse change areas along the riverbanks at the top of the image, QSCDNet is able to accurately detect the subtle changes, which indicates the potential of the quantum features in preserving the change details.

2) Result and Analysis for the Hermiston Dataset: Table V lists the accuracies of the various methods on the Hermiston dataset.

The OAs of all the methods reach over 90%, indicating that the overall performance of these methods on this dataset is better than on the River dataset. This is because the changed areas in the Hermiston dataset are simpler and more organized than those in the River dataset. MFCEN acquires the highest OA, at 98.82%, while the proposed method is in second place, at 98.64%, indicating that the overall classification performance of the proposed method also reaches a good level. For the kappa and $F1$-score, the results are similar to those for the River dataset, and the results of the proposed QSCDNet are the highest, at 95.05% and 96.48%, respectively, which are 0.24% and 0.99% higher than the results of the second-place MFCEN. QSCDNet also obtains the highest precision value, at 96.57%, indicating that the detected changes are relatively accurate and the performance in keeping the boundaries is satisfactory. This is because the changed areas in this dataset are more regular than those in the previous dataset. The highest recall is obtained by QUEEN-G, at 99.0%, while its precision is 90.50%, indicating that it is also relatively balanced in this dataset. The recall of ConvLSTM comes second,

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TABLE V

COMPARATIVE EXPERIMENTAL RESULTS OF THE DIFFERENT HSI-CD METHODS WITH 1% TRAINING SAMPLES ON THE HERMISTON DATASET, WITH THE HIGHEST SCORE HIGHLIGHTED IN BOLD RED, THE SECOND SCORE HIGHLIGHTED IN BOLD GREEN, AND THE THIRD SCORE HIGHLIGHTED IN BOLD BLUE

Method	OA (%)	Kappa (%)	F1 (%)	P (%)	R (%)
ConvLSTM	97.01	88.95	90.75	84.17	**97.47**
CSANet	**98.63**	**93.91**	**94.69**	**94.19**	95.19
SSTFormer	98.60	93.75	94.57	**94.20**	95.19
ML-EDAN	98.62	93.82	94.62	93.75	95.54
MFCEN	**98.82**	**94.81**	**95.49**	94.15	**96.91**
QUEEN-G	98.6	93.7	94.5	90.5	**99.00**
DIEFEN	98.58	93.67	94.48	94.13	95.12
QSCDNet (proposed)	**98.64**	**95.05**	**96.48**	**96.57**	96.41

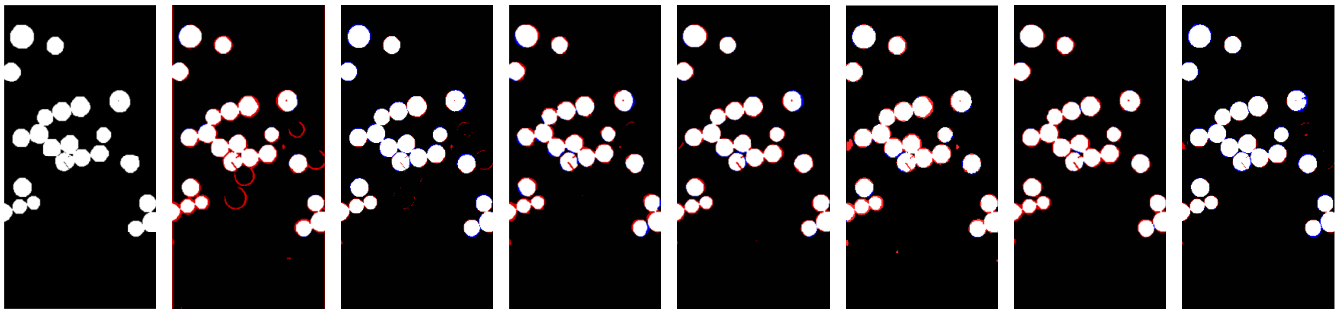


Fig. 4. HSI-CD prediction map of all the methods on the Hermiston dataset. (Left to right) For each row, are the ground truth, the ConvLSTM prediction results, the SSTFormer prediction results, the CSANet prediction results, the MFCEN prediction results, the QUEEN-G prediction results, the DIEFEN prediction results, and the QSCDNet prediction results. The false negative and false positive pixels are denoted in red and blue colors, respectively.

TABLE VI

COMPARATIVE EXPERIMENTAL RESULTS OF THE DIFFERENT HSI-CD METHODS WITH 1% TRAINING SAMPLES ON THE FARMLAND DATASET, WITH THE HIGHEST SCORE HIGHLIGHTED IN BOLD RED, THE SECOND SCORE HIGHLIGHTED IN BOLD GREEN, AND THE THIRD SCORE HIGHLIGHTED IN BOLD BLUE

Method	OA (%)	Kappa (%)	F1 (%)	P (%)	R (%)
ConvLSTM	95.74	89.71	92.51	89.31	**96.66**
CSANet	96.72	92.06	94.38	**93.72**	95.09
SSTFormer	95.41	91.77	93.55	93.17	93.89
ML-EDAN	96.44	91.42	93.94	92.83	95.06
MFCEN	96.78	92.24	94.51	**93.46**	95.62
QUEEN-G	**96.90**	**92.50**	**94.71**	92.62	**97.00**
DIEFEN	**96.86**	**92.48**	**94.70**	92.39	94.68
QSCDNet (proposed)	**97.73**	**94.49**	**96.09**	**95.81**	**96.38**

at 97.47%, while the proposed method is in fourth place, at 96.41%.

The visual results for the Hermiston dataset are shown in Fig. 4. The changed areas in the Hermiston dataset are mainly circular irrigation areas, which are relatively regular. As can be seen from the figure, the main problem with the comparison methods is the false alarms. In addition, there are some missed pixels in the changed areas of SSTFormer and MFCEN in the upper-right corner. The proposed QSCDNet obtains the best performance, with the least blue and red areas, indicating that it can effectively protect the edge contour information while suppressing pseudo-changes.

3) Result and Analysis for the Farmland Dataset: Table VI lists the accuracies of the various methods on the Farmland dataset.

Due to the fact that the changed areas mainly consist of large cultivated land changes, the change detection accuracy for this set of data is also high. Except for ConvLSTM, all the other methods achieve over 90% in kappa. The proposed method obtains the highest OA, kappa, $F1$-score, precision, and recall values, with accuracies of 97.73%, 94.49%, 96.09%, and 95.81%, respectively. For the recall, the highest is QUEEN-G, and the second-place method is ConvLSTM, reaching 96.66%. The proposed QSCDNet obtains third place, at 96.38%.

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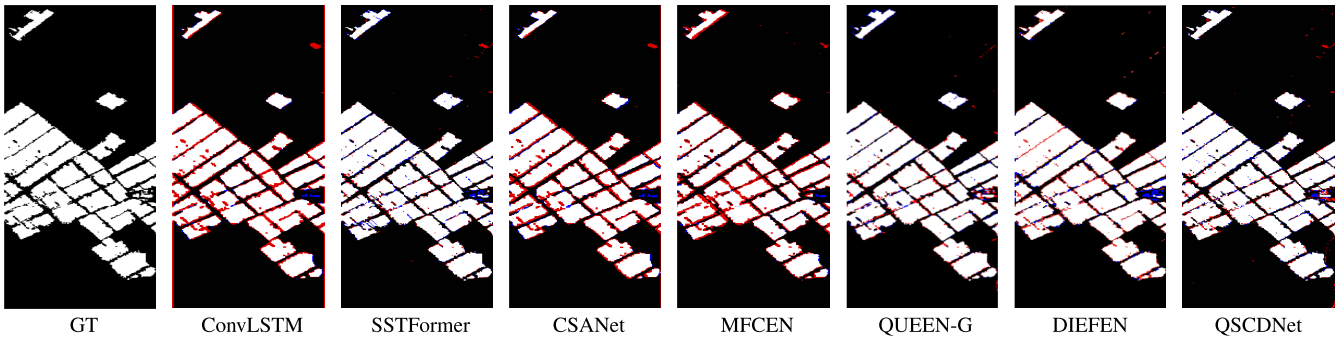


Fig. 5. HSI-CD prediction map of all the methods on the Farmland dataset. (Left to right) For each row, are the ground truth, the QUEEN-G prediction results, the DIEFEN prediction results, and the QSCDNet prediction results. The false negative and false positive pixels are denoted in red and blue colors, respectively.

The visual results for the Farmland dataset are shown in Fig. 5. This dataset mainly focuses on changes in cultivated land, with gaps between the cultivated land being the main challenge. The changed areas may appear regular, but the actual boundary contours are not very smooth. ConvLSTM and CSANet experience significant classification errors at these cultivated land boundaries. SSTFormer and QUEEN-G show the phenomenon of missed detection of farmland boundaries. This corresponds to the accuracy results listed in Table VI. Overall, the proposed QSCDNet method achieves a good balance and the best visual performance.

VI. Conclusion

In this article, a dual-branch hybrid QNN for HSI-CD has been proposed (QSCDNet). The main contribution of this work is that we utilize the quantum state to model potential influencing factors and employ backpropagation optimization to mitigate the effects of pseudo-changes. In addition, in this study, we further analyzed the effectiveness of deeper quantum networks in hyperspectral change detection tasks. Differing from QUEEN-G, the proposed method explores the performance of a deeper quantum network structure for HSI-CD and gives the theoretical modeling of hyperspectral change information based on a quantum network. Furthermore, the proposed QSCDNet network employs a *CRZ* gate to establish the PQC, enabling dynamic adjustment of the relative weights of the quantum superposition states. The experimental results demonstrated that deeper quantum networks are beneficial for hyperspectral change detection. In the future, we plan to conduct further exploration of purely quantum networks.

References

[1] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[2] F. Bovolo and L. Bruzzone, "A split-based approach to unsupervised change detection in large-size multitemporal images: Application to tsunami-damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1658–1670, Jun. 2007.

[3] K. S. Basavaraju, N. Sravya, S. Lal, J. Nalini, C. S. Reddy, and F. Dell'Acqua, "UCDNet: A deep learning model for urban change detection from bi-temporal multispectral Sentinel-2 satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5408110.

[4] M. Chang, X. Meng, W. Sun, G. Yang, and J. Peng, "Collaborative coupled hyperspectral unmixing based subpixel change detection for analyzing coastal wetlands," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8208–8224, 2021.

[5] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Jan. 2002.

[6] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[7] S. Liu, L. Bruzzone, F. Bovolo, and P. Du, "Hierarchical unsupervised change detection in multitemporal hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 244–260, Jan. 2015.

[8] W. Dong, Y. Yang, J. Qu, S. Xiao, and Y. Li, "Local information-enhanced graph-transformer for hyperspectral image change detection with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509814.

[9] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510018.

[10] L. Yang, Y. Chen, S. Song, F. Li, and G. Huang, "Deep Siamese networks based change detection with remote sensing images," *Remote Sens.*, vol. 13, no. 17, p. 3394, Aug. 2021.

[11] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[12] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.

[13] Y. Wu, Z. Bai, Q. Miao, W. Ma, Y. Yang, and M. Gong, "A classified adversarial network for multi-spectral remote sensing image change detection," *Remote Sens.*, vol. 12, no. 13, p. 2098, Jun. 2020.

[14] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.

[15] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529816.

[16] W. Zhang, Y. Zhang, S. Gao, X. Lu, Y. Tang, and S. Liu, "Spectrum-induced transformer-based feature learning for multiple change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500212.

[17] X. Li and J. Ding, "Spectral–temporal transformer for hyperspectral image change detection," *Remote Sens.*, vol. 15, no. 14, p. 3561, Jul. 2023.

[18] Y. Deng, Y. Meng, J. Chen, A. Yue, D. Liu, and J. Chen, "TChange: A hybrid transformer-CNN change detection network," *Remote Sens.*, vol. 15, no. 5, p. 1219, Feb. 2023.

[19] X. Song, Z. Hua, and J. Li, "LHDACT: Lightweight hybrid dual attention CNN and transformer network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 7506005.

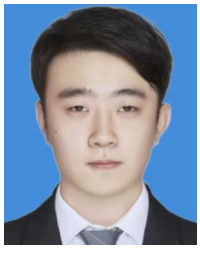
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[20] J. L. Hevia, G. Peterssen, C. Ebert, and M. Piattini, "Quantum computing," *IEEE Softw.*, vol. 38, no. 5, pp. 7–15, Sep. 2021.
[21] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, 2017.
[22] S. Gupta and R. K. P. Zia, "Quantum neural networks," *J. Comput. Syst. Sci.*, vol. 63, no. 3, pp. 355–383, 2001.
[23] I. Cong, S. Choi, and M. D. Lukin, "Quantum convolutional neural networks," *Nature Phys.*, vol. 15, no. 12, pp. 1273–1278, 2019.
[24] G. Chen, Q. Chen, S. Long, W. Zhu, Z. Yuan, and Y. Wu, "Quantum convolutional neural network for image classification," *Pattern Anal. Appl.*, vol. 26, no. 2, pp. 655–667, 2022.
[25] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, "Parameterized quantum circuits as machine learning models," *Quantum Sci. Technol.*, vol. 4, no. 4, Nov. 2019, Art. no. 043001.
[26] A. Sebastianelli, D. A. Zaidenberg, D. Spiller, B. Le Saux, and S. Ullo, "On circuit-based hybrid quantum neural networks for remote sensing imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 565–580, 2022.
[27] A. Chalumuri, R. Kune, S. Kannan, and B. S. Manoj, "Quantum-enhanced deep neural network architecture for image scene classification," *Quantum Inf. Process.*, vol. 20, no. 11, p. 381, Nov. 2021.
[28] F. Fan, Y. Shi, and X. X. Zhu, "Land cover classification from Sentinel-2 images with quantum-classical convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 12477–12489, 2024.
[29] C.-H. Lin and Y.-Y. Chen, "HyperQUEEN: Hyperspectral quantum deep network for image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511720.
[30] C.-H. Lin, T.-H. Lin, and J. Chanussot, "Quantum information-empowered graph neural network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537615.
[31] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.
[32] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.
[33] Y. Lin, S. Li, L. Fang, and P. Ghamisi, "Multispectral change detection with bilinear convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1757–1761, Oct. 2020.
[34] T. Zhan et al., "TDSSC: A three-directions spectral–spatial convolution neural network for hyperspectral image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 377–388, 2021.
[35] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.
[36] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022.
[37] W. Dong et al., "Abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501513.
[38] X. Wang, W. Ni, Y. Feng, and L. Song, "AgF^{2}net: Attention-guided feature fusion network for multitemporal hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[39] W. Zhang, Q. Zhang, H. Ning, and X. Lu, "Cascaded attention-induced difference representation learning for multispectral change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 121, Jul. 2023, Art. no. 103366.
[40] Y. Li et al., "CBANet: An end-to-end cross-band 2-D attention network for hyperspectral change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513011.
[41] X. Ou, L. Liu, B. Tu, L. Qing, G. Zhang, and Z. Liang, "CBW-MSSANet: A CNN framework with compact band weighting and multiscale spatial attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507618.
[42] X. Wang, K. Zhao, X. Zhao, and S. Li, "GTMSiam: Gated transmitting-based multiscale Siamese network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023.
[43] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[44] J. Ding, X. Li, and L. Zhao, "CDFormer: A hyperspectral image change detection method based on transformer encoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6015405.
[45] L. Wu et al., "DIEFEN: Differential information-enhanced feature exchange network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5533312.
[46] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.
[47] W. Zhang, L. Su, Y. Zhang, and X. Lu, "A spectrum-aware transformer network for change detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518612.
[48] X. Zhang, S. Tian, G. Wang, X. Tang, J. Feng, and L. Jiao, "CAST: A cascade spectral-aware transformer for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520714.
[49] J. Ding, X. Li, S. Xiang, and S. Chen, "Multilevel features fused and change information enhanced neural network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502413.
[50] Z. Chen et al., "TL2GH^{2}T: Triple-path local-to-global network with hybrid head transformer for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528112.
[51] S. Oh, J. Choi, and J. Kim, "A tutorial on quantum convolutional neural networks (QCNN)," in *Proc. Int. Conf. Inf. Commun. Technol. Converg. (ICTC)*, 2020, pp. 236–239.
[52] Y. Li, R.-G. Zhou, R. Xu, J. Luo, and W. Hu, "A quantum deep convolutional neural network for image recognition," *Quantum Sci. Technol.*, vol. 5, no. 4, Jul. 2020, Art. no. 044003.
[53] D. Bokhan, A. S. Mastiukova, A. S. Boev, D. N. Trubnikov, and A. K. Fedorov, "Multiclass classification using quantum convolutional neural networks with hybrid quantum-classical learning," *Frontiers Phys.*, vol. 10, Nov. 2022, Art. no. 1069985.
[54] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2018, pp. 7132–7141.
[55] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge, U.K.: Cambridge Univ. Press, 2010.
[56] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, "Quantum circuit learning," *Phys. Rev. A, Gen. Phys.*, vol. 98, no. 3, 2018, Art. no. 032309.
[57] S. Liu, Q. Du, X. Tong, A. Samat, H. Pan, and X. Ma, "Band selection-based dimensionality reduction for change detection in multi-temporal hyperspectral images," *Remote Sens.*, vol. 9, no. 10, p. 1008, Sep. 2017.
[58] P. Henry, G. Chander, B. Fougnie, C. Thomas, and X. Xiong, "Assessment of spectral band impact on intercalibration over desert sites using simulation based on EO-1 hyperion data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1297–1308, Mar. 2013.



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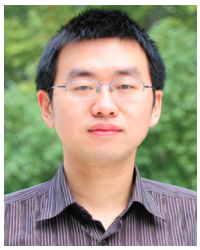
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